Microbial co-occurrence network reveals climate and geographic patterns for soil diversity on the planet

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Highlights

- **A network perspective** to better understand **how climatic and geographic factors affect co-occurrence patterns of soil microbiomes**.
- Overview of **soil microbial co-occurrence patterns in relation to environment-specific microbial ecosystems**.

Background

- Limited systematic investigations on how microbial communities are affected by geographic and climatic factors.
- Exploration of microbial diversity in specific geographic areas with focus on the underlying microbial interactions rather than the structure changes.

Aim Of The Study

Explore the effects of key climatic factors and geographic patterns on the soil diversity and microbial interactions across the globe.

Methodology

1. *Earth Microbiome Project*[1] (EMP) a massive collection of planetary-scale microbiome datasets.
2. *Köppen-Geiger climate classification*[2] along with bioclimatic variables from the *WorldClim* database.
3. A microbial co-occurrence network was built to capture soil microbial interactions using the *SpiecEasi*[3] workflow.

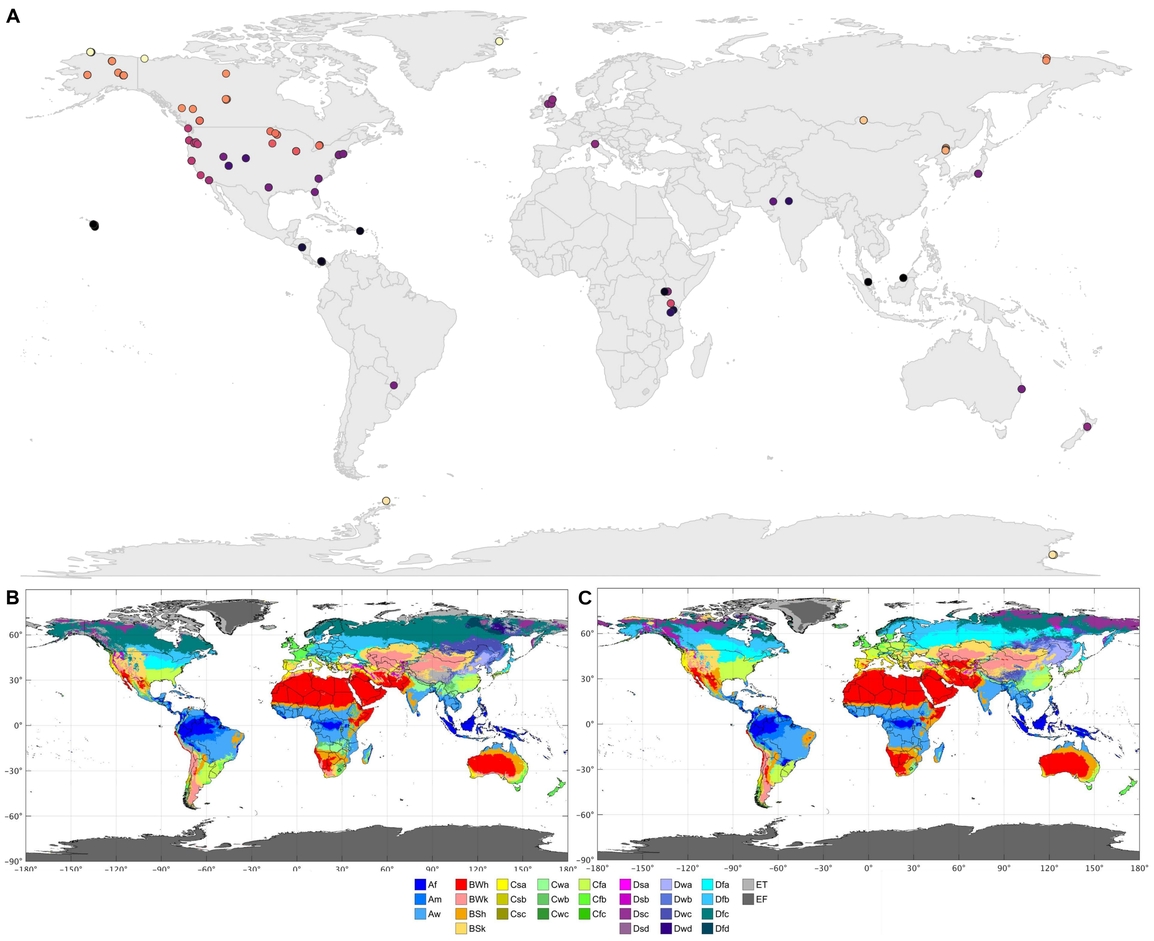

Fig 1. (**A**) A subset of 364 soil (non-saline) samples was used to study the soil diversity on the planet. (**B**) Present-day map (1980–2016) and (**C**) future map (2071–2100) of Köppen-Geiger climate classification maps at 1-km resolution.

Results

Alpha diversity tends to follow different distributions in Arid and Polar environments depending on Precipitation type and Level of Heat.

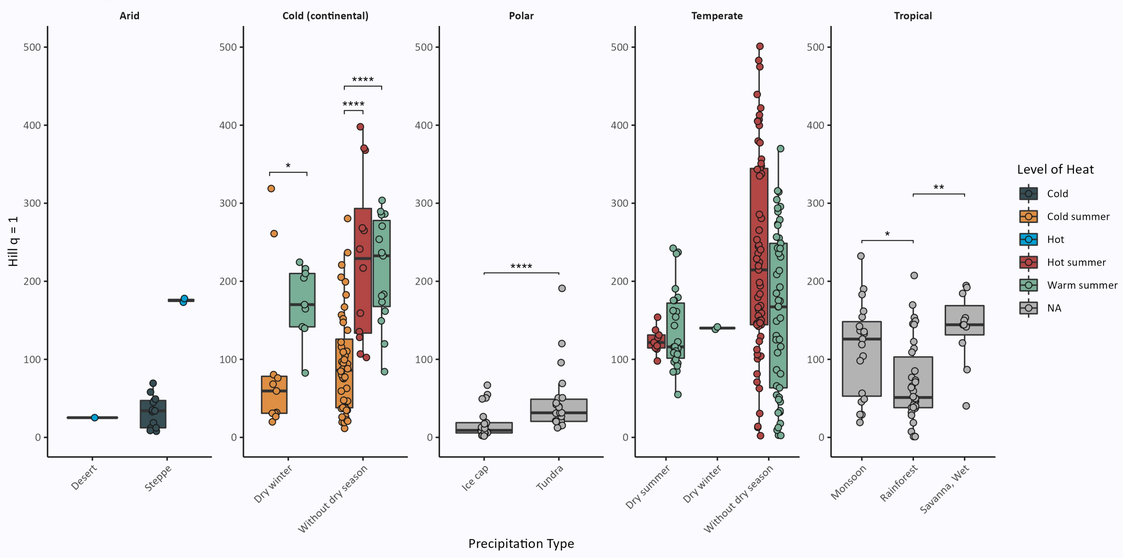

Fig 2. Boxplots of Hill number ($q = 1$) distributions across different precipitation types and level of heat in the environments. Wilcoxon pairwise comparisons were used to assess significance between the distributions. For each comparison, p values after FDR adjustment are considered.

A complete network of **3050 tag-sequences (taxa nodes)** and **67,664 edges** was built using the *SpeacEasi* algorithm (Meinshausen and Bühlmann neighborhood selection framework).

15 sub-networks *were generated by keeping tag sequences present in specific samples (eventually* ***in specific environments****) and all edges among them in the co-occurrence network.*

Every sequence *is* ***present in at least 5 different climate types****.*

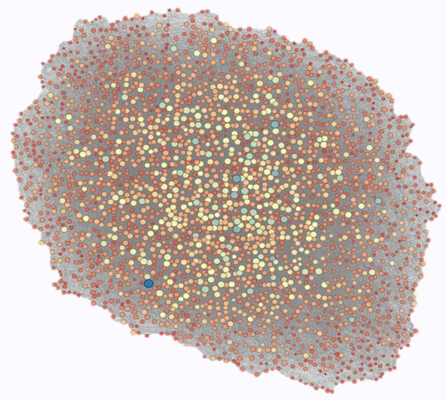

Fig 3. Overall network produced for the 364 soil samples from the Earth Microbiome Project.

Arid and Polar environments (e.g. samples) contain less than 50% of the observed tag-sequences, while preserving strong co-expression patterns.

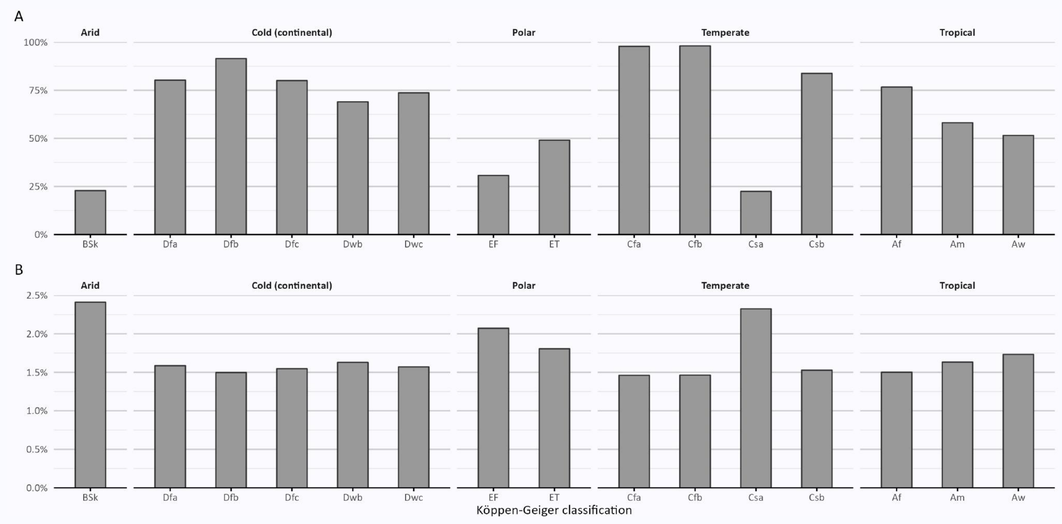

Fig 4. General topological statistics of the constructed sub-networks (**A**) Percentage of tag-sequences present in every environment (sub-network) in relation to the total number of tag-sequences present in the overall network. (**B**) Percentage of connections (co-expression) in every environment (sub-network) in relation to the possible number of connections in every sub-network.

Increased degree centrality scores were observed in BSk (Cold semi-arid – steppe), Dwb (Monsoon-influenced warm-summer humid continental) and Csa (Hot-summer Mediterranean) environments.

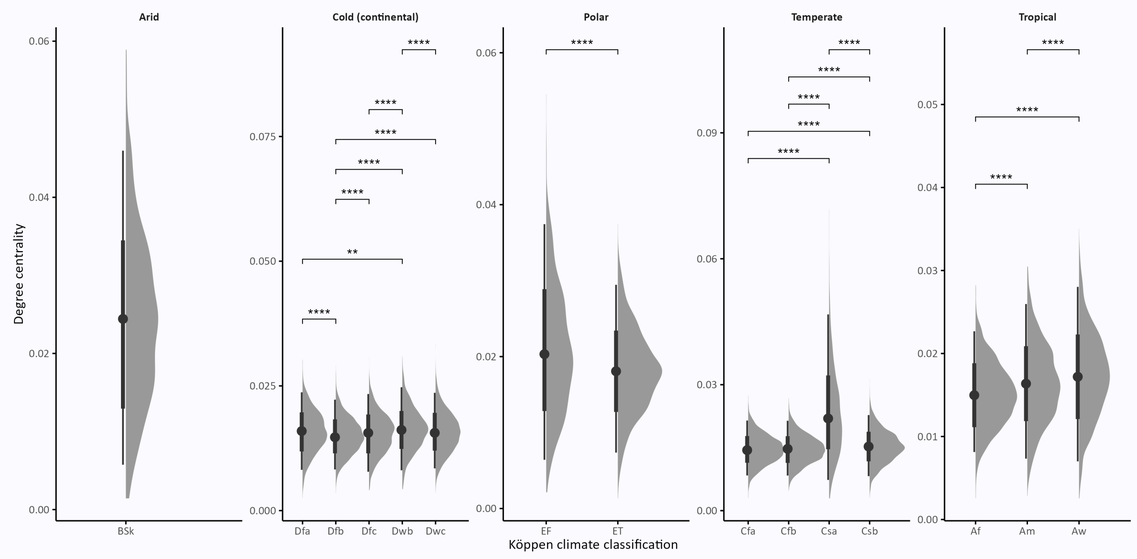

Fig 5. Degree centrality distributions across different Köppen-Geiger climate classifications. Wilcoxon pairwise comparisons were used to assess significance between the distributions. For each comparison, p values after FDR adjustment are considered.

CONCLUSIONS

- Identify and prioritize **climatic and geographic patterns for soil diversity on a global scale**.
- **Interconnection patterns** across microbiomes **in extreme climatic environments** (Arid, Polar – BSk, Dwb, Csa).
- **Significant topological shifts** (degree centrality distribution) **on network level** for different climatic regions.
- The application of this strategy could be used to **recognize key-factors for environment-specific or global attributes of microbial ecosystems**.

References

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2. *Beck, H. et al. Present and future Köppen-Geiger climate classification maps at 1-km resolution. Sci Data 5, 180214. https://doi.org/10.1038/sdata.2018.214*
3. *Kurtz ZD et al. Sparse and Compositionally Robust Inference of Microbial Ecological Networks. PLOS Computational Biology 11(5). https://doi.org/10.1371/journal.pcbi.1004226*

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